UTA013:ENGINEERINGDESIGNPROJECT-I (2 Self Hours)				
	L	T	P	Cr
	1	0	2	3.0

Course Objective:The project will introduce students to the challenge of electronic systems design & integration. The project is an example of 'hardware and software co-design' and the scale of the task is such that it will require teamwork as a coordinated effort.

Syllabus

To provide a basis for the technical aspects of the project a small number of lectures are incorporated into the module. As the students would have received little in the way of formal engineering instruction at this early stage in the degree course, the level of the lectures is to be introductory with an emphasis on the physical aspects of the subject matter as applied to the 'Mangonel' project. The lecture series include subject areas such as Materials, Structures, Dynamics and Digital Electronics delivered by experts in the field.

Thismoduleisdeliveredusingacombinationofintroductorylecturesandparticipationby the students in 15"activities". The activities are executed to support the syllabus of the course and might take place in specialised laboratories or on the open ground used for firing the Mangonel. Students work in groups throughout the semester to encourage teamwork, cooperation and to avail of the different skills of its members. In the end the students work in sub-groups to do the Mangonel throwing arm redesign project. They assemble and operate a Mangonel, based on the lectures and tutorials assignments of mechanical engineering they experiment with the working, critically analyse the effect of design changes and implement the final project in a competition. Presentation of the group assembly, redesign and individual reflection of the project is assessed in the end.

Break up of lecture details to be taken up by MED:

LecNo.	Topic	Contents
Lec1	Introduction	The Mangonel Project. History. Spreadsheet.
Lec2	PROJECTILE MOTION	no DRAG, Design spread sheet simulator for it.
Lec3	PROJECTILE MOTION	with DRAG, Design spread sheet simulator for it.
Lec4	STRUCTURES FAILURE	STATIC LOADS
Lec5	STRUCTURES FAILURE	DYNAMIC LOADS
Lec6	REDESIGNING THE MANGONEL	Design constraints and limitations of materials for redesigning the Mangonel for competition as a group.
Lec7	MANUFACTURING	Manufacturing and assembling the Mangonel.
Lec8	SIMULATION IN ENGINEERING DESIGN	Simulation as an Analysis Tool in Engineering Design.

Lec9	ROLE OF MODELLING & PROTOTYPING	The Role of Modelling in Engineering Design.

Breakup of lecture details to be taken up by ECED:

LecNo.	Topic	Contents
Lec1-5	Digital Electronics	Prototype, Architecture, Using the integrated development Environment(IDE)to prepare an arduino sketch, structuring an arduino program, using simple primitive types(Variables), Simple programming examples. Definition of a sensor and actuator.

Laboratory Work:

AssociatedLaboratory/ProjectProgram:T-MechanicalTutorial,L-ElectronicsLaboratory, W-MechanicalWorkshopof“Mangonel”assembly,redesign,operationandreflection.

Titlefortheweeklyworkin15weeks	Code
Using a spread sheet to develop a simulator	T1
Dynamics of projectile launched by a Mangonel-No Drag	T2
Dynamics of projectile launched by a Mangonel-With Drag	T3
Design against failure under staticactions	T4
Design against failure under dynamicactions	T5
Electronics hardware and Arduino controller	L1
Electronics hardware and Arduino controller	L2
Programming the Arduino Controller	L3
Programming the Arduino Controller	L4
Final project of sensors, electronics hardware and programmed Arduino controller-based measurement of angular velocity of the“Mangonel”throwing arm.	L5
Assembly of the Mangonel by group	W1
Assembly of the Mangonel by group	W2
Innovative redesign of the Mangonel and its testing by group	W3
Innovative redesign of the Mangonel and its testing by group	W4
Final intergroup competition to assess best redesign and understanding of the“Mangonel”.	W5

Project: TheProjectwillfacilitatethedesign,constructionandanalysisofa “Mangonel”.In additiontosomeintroductorylectures,thecontentofthestudents’workduringthesemesterwillconsistof:

1. the assembly of a Mangonel from a Bill Of Materials (BOM), detailed engineering drawings of parts, assembly instructions, and few pre-fabricated parts;
2. The development of a software tool to allow the trajectory of a“missile” to be studied as a function of various operating parameters in conditions of no-drag and drag due to air;